RAK19010 WisBlock Base Board with Power Slot Datasheet

Overview

Description

RAK19010 is a **WisBlock Base Board with Power Slot** that connects **WisBlock Core** and other **WisBlock Modules**. The power slot of RAK19010 is required to have an attached WisBlock Power Slot module that provides power supply to the core and other modules. There are many different types of power slot modules compatible with RAK19010 and the choice will depend on the type of application.

It has one slot reserved for the power slot module, one for the core module, one slot for the IO module, and four sensor slots A-D for small WisBlock modules. The WisBlock Core, Power, and IO modules are attached on the top side, and smaller WisBlock modules can be attached to the top or bottom side of the RAK19010. Slot A and D hold modules up to 23 mm in size, while all slots A up to D support 10 mm WisBlock modules. Also, there are three **2.54 mm pitch headers** for extension interface with **BOOT**, **GPIO**, **ADC**, **I2C**, and **UART** pins.

WisBlock modules are connected to the RAK19010 WisBlock Base board via high-speed board-to-board connectors. They provide secure and reliable interconnection to ensure the signal integrity of each data bus. A set of screws is used to fix the modules, making it reliable even in an environment with lots of vibrations.

You can also use a RAK19005 WisBlock Sensor Extension Cable or RAK19008 WisBlock IO Extension Cable to position the WisBlock modules apart from the WisBlock Base board or in any part of your case.

Applications

- Wireless sensor network
- Environmental monitoring
- Wireless data transmission
- Data acquisition in the industrial environment
- Location and tracking of personnel or moving objects

Features

- Flexible building block design, which enables modular function realization and expansion
- High-speed interconnection secured with screws to ensure signal integrity
- Supports multiple types of low-power MCUs
- Supports multiple types of sensors - a single board can support a combination of two different types of sensors
- Supports different power modules depending on the applications.
    - RAK19012 - USB, LiPo and Solar
    - RAK19013 - LiPo and Solar
    - RAK19015 - Battery
    - RAK19016 - 5 V to 24 V voltage input

- **Module Slots**
  - 1 WisBlock Core module
  - 1 WisBlock power module
  - 1 WisBlock module compatible with IO slot
  - 4 WisBlock modules compatible with slots A-D
  - Pin headers accessible pins for BOOT, GPIO, ADC, I2C, and UART interfaces
- **Size**
  - **RAK19010** has a size of only 30 × 60 mm, which lets you create solutions that fit into the smallest housings.

If you can't find a WisBlock module that fits your IoT requirements, use the standard connectors of WisBlock to develop your specific function module. WisBlock supports open-source hardware architecture and you can find tutorials showing how to create your own Awesome WisBlock module.

Specifications

Block Overview

There are seven (7) slots on RAK19010 WisBlock Base Board with Power Slot:

- **CPU SLOT**: This slot is reserved for the WisBlock Core module which has the main MCU.
- **Power SLOT**: This slot is required to provide power to WisBlock Core and modules.
- **IO SLOT**: This slot is used for IO extension modules.
- **Four Sensor Slots:** The sensor slots A to D are used to connect with the I2C bus. Slots A and D can be used for GNSS modules, too.

Also, there are pin headers 2.54 mm pitch for the extension interface that connects to BOOT, I2C, UART, ADC, and GPIO pins.

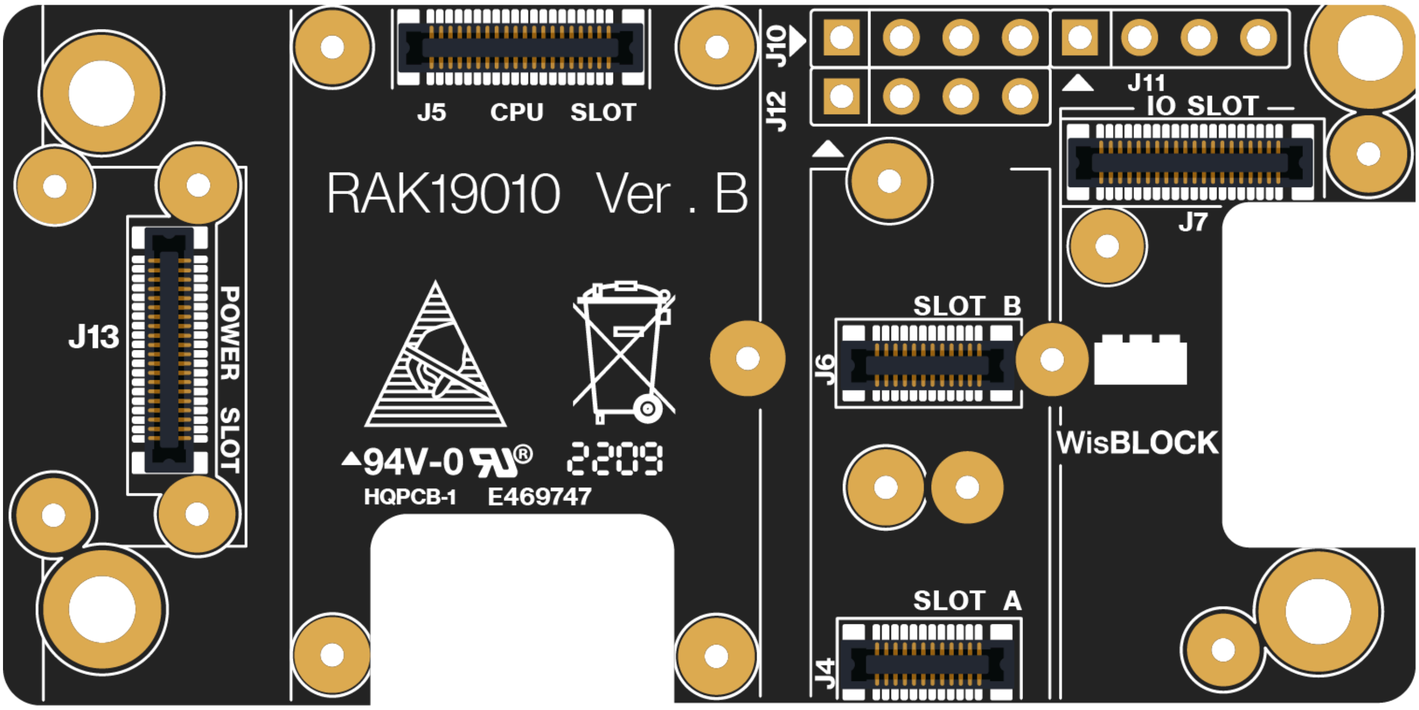

Figure 1: RAK19010 WisBlock Base Board with Power Slot top view

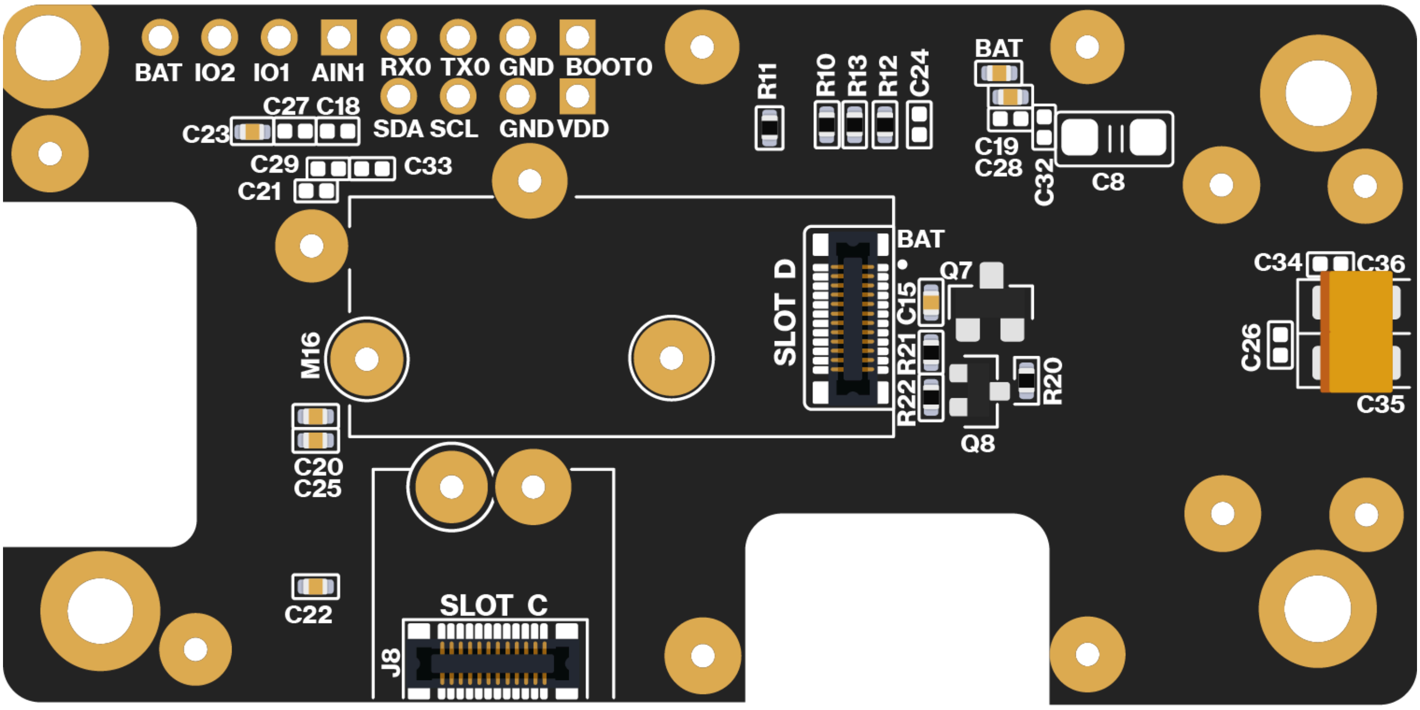

Figure 2: RAK19010 WisBlock Base Board with Power Slot bottom view

Block Diagram

The block diagram in **Figure 3** shows the internal architecture and external interfaces of the RAK19010 board.

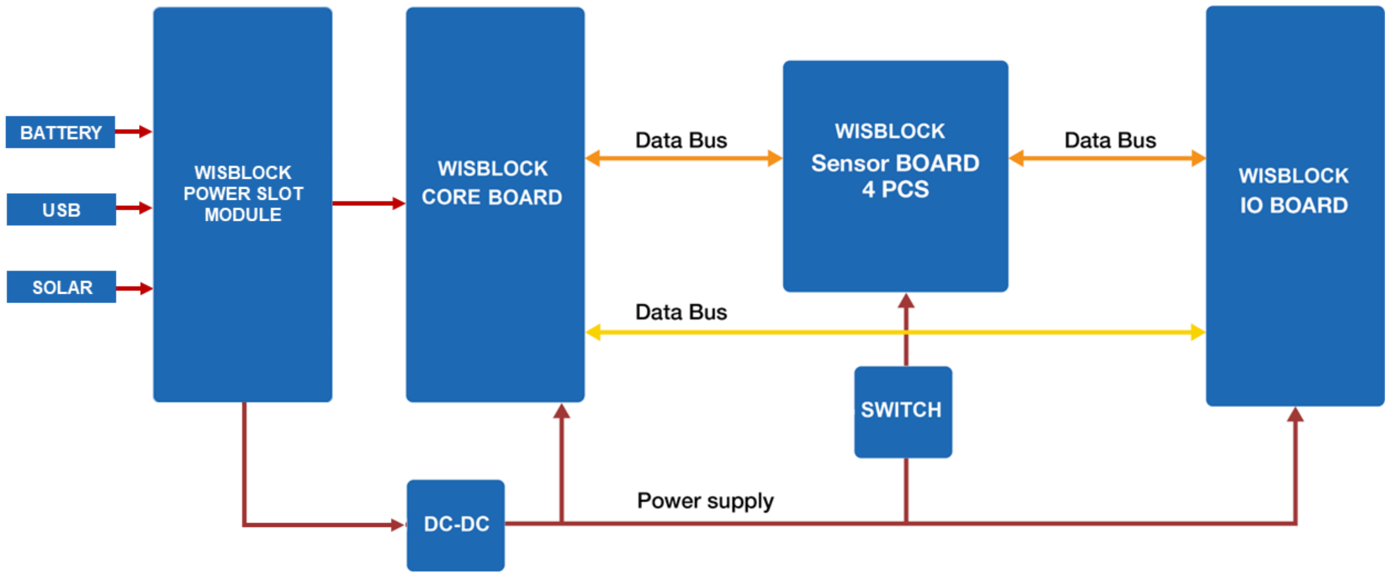

Figure 3: RAK19010 WisBlock Base Board wth Power Slot block diagram

Hardware

The hardware specification is categorized into six parts. It shows the interfacing, pinouts, and their corresponding functions and diagrams. It also presents the electrical, environmental, and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK19010 WisBlock Base Board 2nd Gen.

Interfaces

RAK19010 WisBlock Base Board with Power Slot provides the following interfaces, headers, a button, and WisBlock Connectors.

- One connector for CPU slot
- One connector for the power slot
- One connector for the IO slot
- Four connectors for WisBlock sensor modules (slots A to D)
- Three 4-pin header 2.54 mm hole pads (GPIO, ADC, UART, I2C, Power)

Figure 4 show the location of RAK19010 main components.

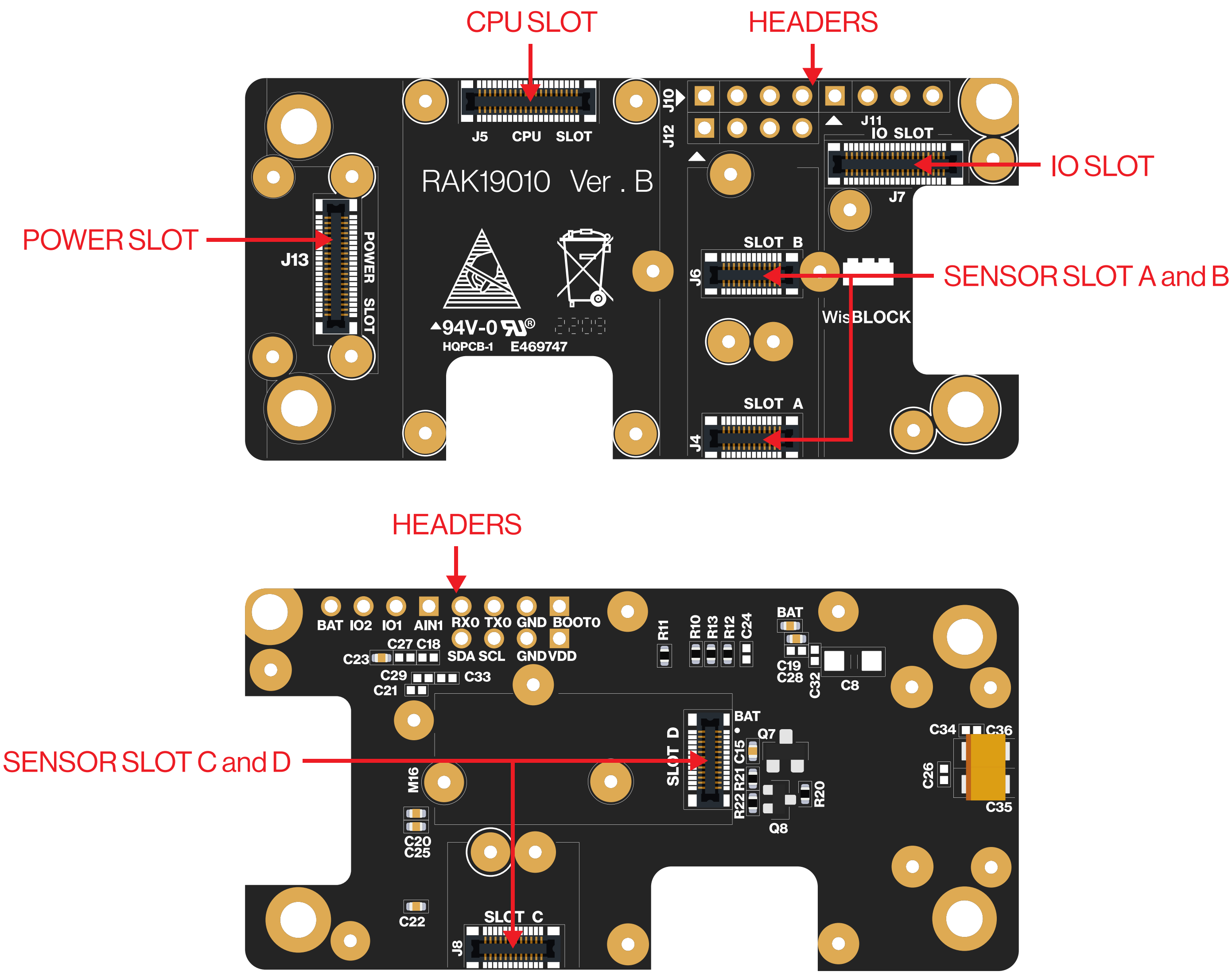

Figure 4: RAK19010 top and bottom parts

J10, J11, J12 Headers

On the RAK19010 Base Board, there are three 2.54 mm pitch headers for IO extension. BOOT, I2C, GPIO, and UART pins from the WisBlock Core module are also connected to these headers.

J10 Header Pinout

Pin	Pin Name	Description
1	BOOT	MCU BOOT pin
2	GND	Ground pin
3	TX1	UART1 TX pin
4	RX1	UART1 RX pin

J11 Header Pinout

Pin	Pin Name	Description
1	AIN1	ADC input signal
2	IO1	General purpose IO
3	IO2	General purpose IO
4	VBAT	Battery voltage

J12 Header Pinout

Pin	Pin Name	Description
1	VDD	3.3 V
2	GND	Ground pin
3	SCL	I2C1 clock
4	SDA	I2C2 data

NOTE

BOOT pin is used on startup configuration or sequence of the WisBlock Core connected to it. It is commonly used for uploading the bootloader and/or application firmware. The requirements of the state of the BOOT pin depend on the specific model of the WisBlock Core used.

Pin Definition

Connector for WisBlock Core

The **WisBlock Core module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

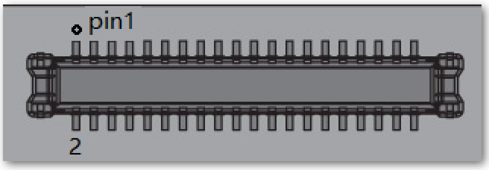

Figure 5: WisBlock Core module connector

The table below shows the pinout of the 40-pin WisBlock core connector:

Function Name of WisBlock Base	Pin Number	Pin Number	Function Name of WisBlock Base
VBAT	1	2	VBAT
GND	3	4	GND
3V3	5	6	3V3
USB+	7	8	USB–
VBUS	9	10	SW1
TXD0	11	12	RXD0
RESET	13	14	LED1
LED2	15	16	LED3
VDD	17	18	VDD
I2C1_SDA	19	20	I2C1_SCL
AIN0	21	22	AIN1
BOOT0	23	24	IO7
SPI_CS	25	26	SPI_CLK

Function Name of WisBlock Base	Pin Number	Pin Number	Function Name of WisBlock Base
SPI_MISO	27	28	SPI_MOSI
IO1	29	30	IO2
IO3	31	32	IO4
TXD1	33	34	RXD1
I2C2_SDA	35	36	I2C2_SCL
IO5	37	38	IO6
GND	39	40	GND

As for the following table, it shows the definition of each pin of the WisBlock Core connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	VBUS for USB
10	SW1	I/O	Switch signal for customer's control
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset

Pin Number	Pin Name	Type	Description
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	LED3	I/O	LED for custom usage
17	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
18	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signals
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	BOOT0	I	For ST MCU, set high when reset. The MCU will allow you to enter boot mode.
24	IO7	I/O	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 RX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal

Pin Number	Pin Name	Type	Description
36	I2C2_SCL	I/O	The second set of I2C data signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connector for WisBlock Power

The **WisPower module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

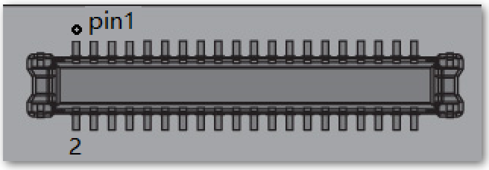

Figure 6: WisBlock Power Slot module connector

The table below shows the pinout of the 40-pin power module connector:

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
1	VBAT	2	VBAT
3	GND	4	GND
5	3V3	6	3V3
7	USB+	8	USB–
9	VBUS	10	VBUS
11	NC	12	NC
13	RESET	14	LED1
15	LED2	16	NC
17	3V3	18	3V3
19	I2C1_SDA	20	I2C1_SCL

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
21	AIN0	22	AIN1
23	NC	24	NC
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	NC	30	NC
31	NC	32	NC
33	NC	34	NC
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND	40	GND

As for the following table, it shows the definition of each pin of the WisBlock Power connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	USB VBUS
10	VBUS	S	USB VBUS

Pin Number	Pin Name	Type	Description
11	NC	NC	Not connected
12	NC	NC	Not connected
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	NC	NC	Not connected
17	3V3	S	3.3 V power supply
18	3V3	S	3.3 V power supply
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signal
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connected
24	NC	NC	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock signal
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	NC	NC	Not connected
30	NC	NC	Not connected
31	NC	NC	Not connected
32	NC	NC	Not connected

Pin Number	Pin Name	Type	Description
33	NC	NC	Not connected
34	NC	NC	Not connected
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connectors for WisBlock Sensor

The WisBlock sensor module connector is a **24-pin board-to-board connector**.

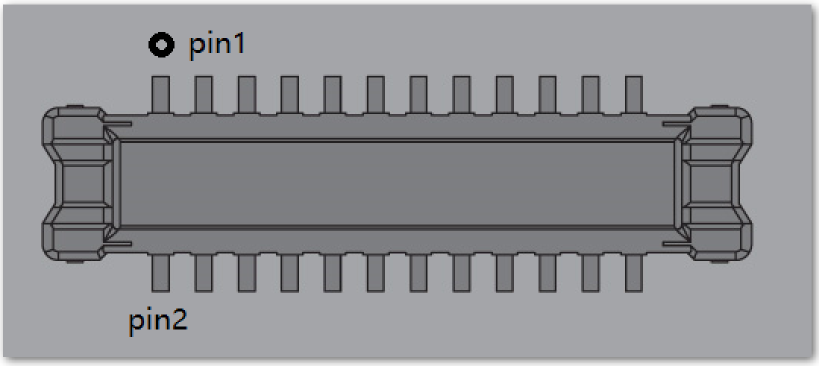

Figure 7: WisBlock Sensor module connector

> **NOTE**
> There are four connectors reserved for the sensor modules on the RAK19010. The pinout definition of the WisBlock modules with a 24-pin connector on WisBlock Base Board varies according to its connector.

Connector D	Connector C	Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B	Connector C	Connector D
TXD1	NC	NC	TXD1	1	2	GND	GND	GND	GND
SPI_CS	SPI_CS	SPI_CS	SPI_CS	3	4	SPI_CLK	SPI_CLK	SPI_CLK	SPI_CLK

Connector D	Connector C	Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B	Connector C	Connector D
SPI_MISO	SPI_MISO	SPI_MISO	SPI_MISO	5	6	SPI_MOSI	SPI_MOSI	SPI_MOSI	SPI_MOSI
I2C1_SCL	I2C1_SCL	I2C1_SCL	I2C1_SCL	7	8	I2C1_SDA	I2C1_SDA	I2C1_SDA	I2C1_SDA
VDD	VDD	VDD	VDD	9	10	IO2	IO1	IO4	IO6
3V3_S	3V3_S	3V3_S	3V3_S	11	12	IO1	IO2	IO3	IO5
NC	NC	NC	NC	13	14	3V3_S	3V3_S	3V3_S	3V3_S
NC	NC	NC	NC	15	16	VDD	VDD	VDD	VDD
NC	NC	NC	NC	17	18	NC	NC	NC	NC
NC	NC	NC	NC	19	20	NC	NC	NC	NC
NC	NC	NC	NC	21	22	NC	NC	NC	NC
GND	GND	GND	GND	23	24	RXD1	NC	NC	RXD1

As for the following table, it shows the pin name and description of each pin in the WisBlock Sensor module connector.

Pin Number	Connector A	Connector B	Connector C	Connector D	Type	Description
1	TXD1	NC	NC	TXD1	I/O	UART TX signal
2	GND	GND	GND	GND	S	Ground
3	SPI_CS	SPI_CS	SPI_CS	SPI_CS	I/O	SPI chip select signal
4	SPI_CLK	SPI_CLK	SPI_CLK	SPI_CLK	I/O	SPI clock
5	SPI_MISO	SPI_MISO	SPI_MISO	SPI_MISO	I/O	SPI MISO signal
6	SPI_MOSI	SPI_MOSI	SPI_MOSI	SPI_MOSI	I/O	SPI MOSI signal
7	I2C1_SCL	I2C1_SCL	I2C1_SCL	I2C1_SCL	I/O	I2C clock signal
8	I2C1_SDA	I2C1_SDA	I2C1_SDA	I2C1_SDA	I/O	I2C data signal

Pin Number	Connector A	Connector B	Connector C	Connector D	Type	Description
9	VDD	VDD	VDD	VDD	S	Generated by CPU module. Used to power sensor board if MCU IO level is not 3.3 V
10	IO2	IO1	IO4	IO6	I/O	General purpose IO. IO2 controls the power switch of 3V3_S. When the 3V3_S function is used, IO2 can not be used as an interrupt of the sensor.
11	3V3_S	3V3_S	3V3_S	3V3_S	S	3.3 V power supply. Can be shut down by the CPU module.
12	IO1	IO2	IO3	IO5	I/O	General purpose IO - IO controls the power switch of 3V3_S. When the 3V3_S function is used, IO2 cannot be used as an interrupt of the sensor.
13	NC	NC	NC	NC	NC	Not connected
14	3V3_S	3V3_S	3V3_S	3V3_S	S	3.3 V power supply. Can be shut down by the CPU module.
15	NC	NC	NC	NC	NC	Not connected
16	VDD	VDD	VDD	VDD	S	Generated by CPU module. Used to power sensor board if the MCU IO level is not 3.3 V.
17	NC	NC	NC	NC	NC	Not connected
18	NC	NC	NC	NC	NC	Not connected
19	NC	NC	NC	NC	NC	Not connected
20	NC	NC	NC	NC	NC	Not connected
21	NC	NC	NC	NC	NC	Not connected
22	NC	NC	NC	NC	NC	Not connected
23	GND	GND	GND	GND	S	Ground
24	RXD1	NC	NC	RXD1	I/O	UART RX signal

Connector for WisBlock IO Slot

The WisBlock Module IO Slot connector, as shown in **Figure 8**, is a 40-pin board-to-board connector.

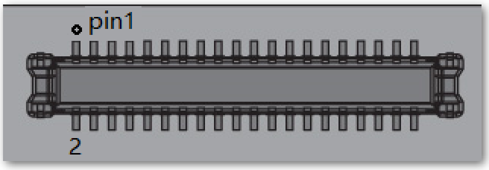

Figure 8: WisBlock IO slot connector

Pinout definition for IO slot:

Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B
VBAT	VBAT	1	2	VBAT	VBAT
GND	GND	3	4	GND	GND
3V3	3V3	5	6	3V3_S	3V3_S
USB+	USB+	7	8	USB–	USB–
VBUS	VBUS	9	10	SW1	SW1
TXD0	TXD0	11	12	RXD0	RXD0
RESET	RESET	13	14	LED1	LED1
LED2	LED2	15	16	LED3	LED3
VDD	VDD	17	18	VDD	VDD
I2C1_SDA	I2C1_SDA	19	20	I2C1_SCL	I2C1_SCL
AIN0	AIN0	21	22	AIN1	AIN1
NC	NC	23	24	NC	NC
SPI_CS	SPI_CS	25	26	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	27	28	SPI_MOSI	SPI_MOSI
IO1	IO1	29	30	IO2	IO2
IO3	IO3	31	32	IO4	IO4
TXD1	TXD1	33	34	RXD1	RXD1

Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B
I2C2_SDA	I2C2_SDA	35	36	I2C2_SCL	I2C2_SCL
IO5	IO5	37	38	IO6	IO6
GND	GND	39	40	GND	GND

As for the following table, it shows the pin name and description of the WisBlock IO module connector.

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3_S	S	3.3 V power supply. Can be shut down by a CPU module.
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	5 V input for USB
10	SW1	I/O	Switch signal for custom used
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charge indicator
15	LED2	I/O	LED for custom used
16	LED3	I/O	LED for custom used
17	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V

Pin Number	Pin Name	Type	Description
18	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V.
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signals
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connect
24	NC	NC	Not connect
25	SPI_CS	I/O	SPI chip select signal
26	SP_CLK	I/O	SPI clock
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 TX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground

Pin Number	Pin Name	Type	Description
40	GND	S	Ground

Electrical Characteristics

Absolute Maximum Ratings

The Absolute Maximum Ratings of the device are shown in the table below. The stress ratings are the functional operation of the device.

> **WARNING**
> 1. If the stress rating goes above what is listed, it may cause permanent damage to the device.
> 2. Under the listed conditions is not advised.
> 3. Exposure to maximum rating conditions may affect the device reliability.

Ratings	Maximum Value	Unit
IOs of WisConnector	−0.3 to VDD+0.3	V
ESD	2000	V

> **WARNING**
> The RAK19010, like any electronic equipment, is sensitive to **electrostatic discharge (ESD)**. Improper handling can cause permanent damage to the module.

Mechanical Characteristics

Board Dimensions

> **NOTE**
> You may also refer and download the [M1.2 Stand-off fastener/inserts datasheet](#) .

Figure 9 shows the detailed mechanical dimensions of the RAK19010 Board.

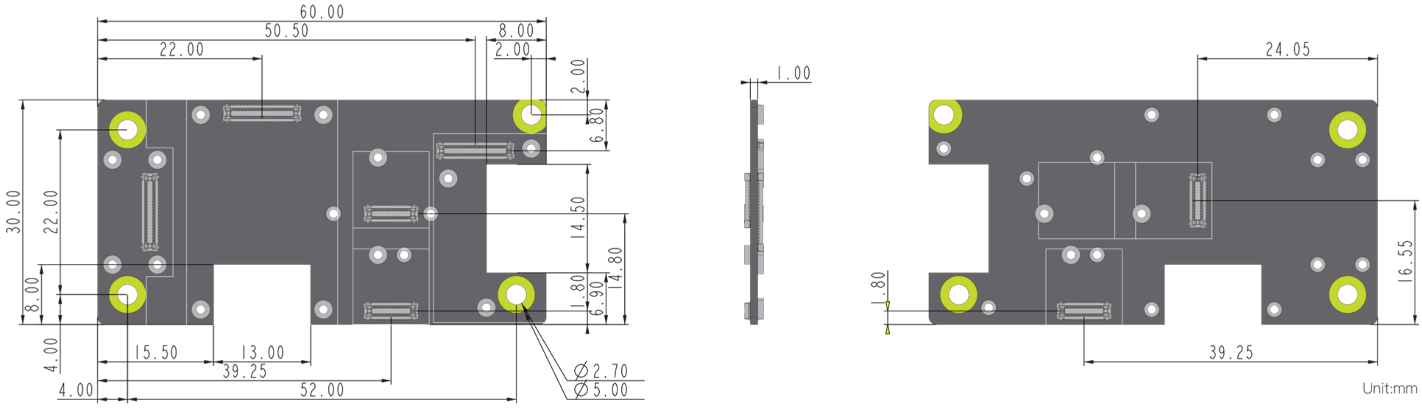

Figure 9: RAK19010 mechanical board dimension

Figure 10 shows the mounting holes location and diameter of the RAK19010 Board.

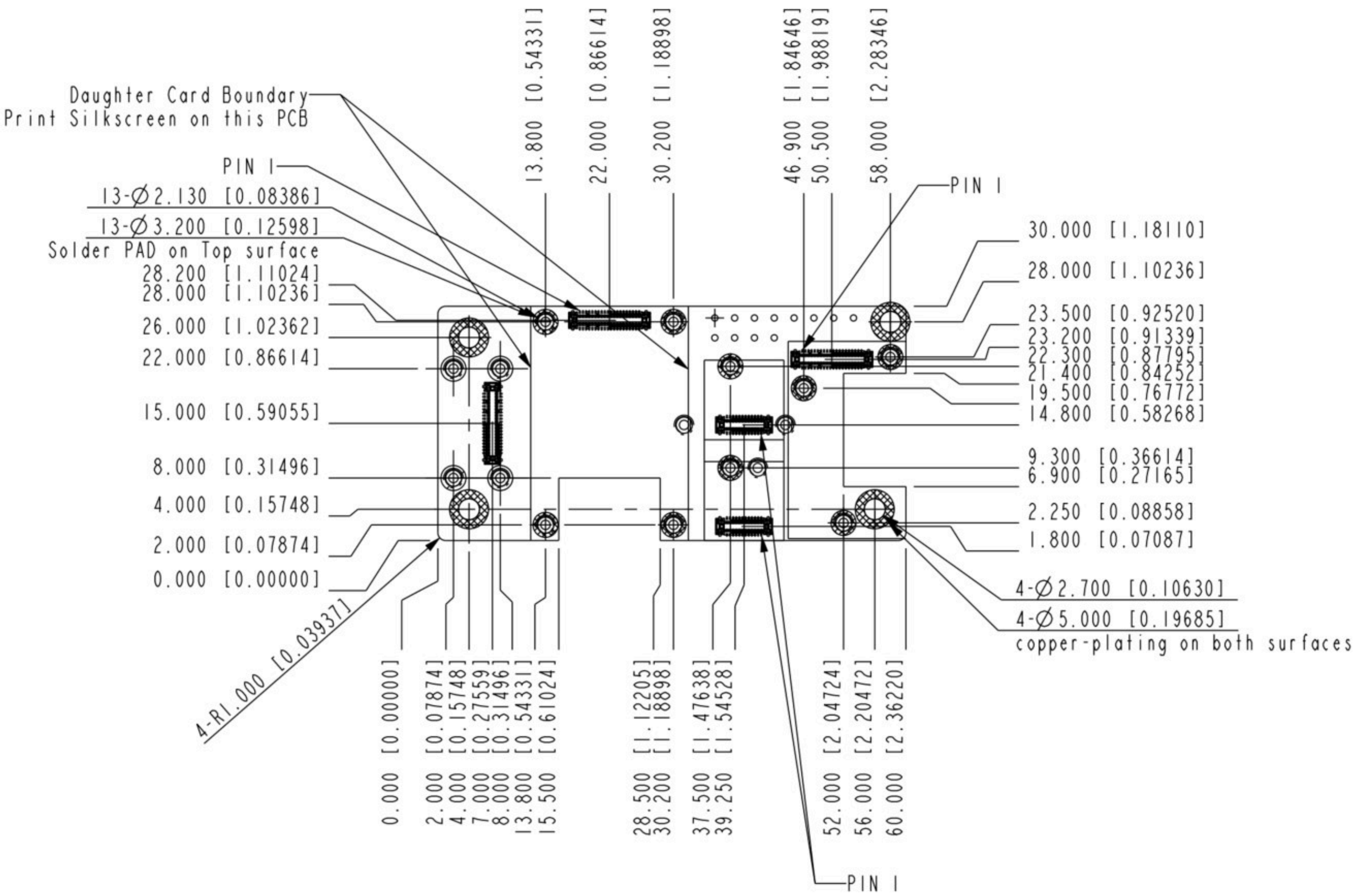

Figure 10: RAK19010 mechanical dimensions and mounting holes locations

WisConnector PCB Layout

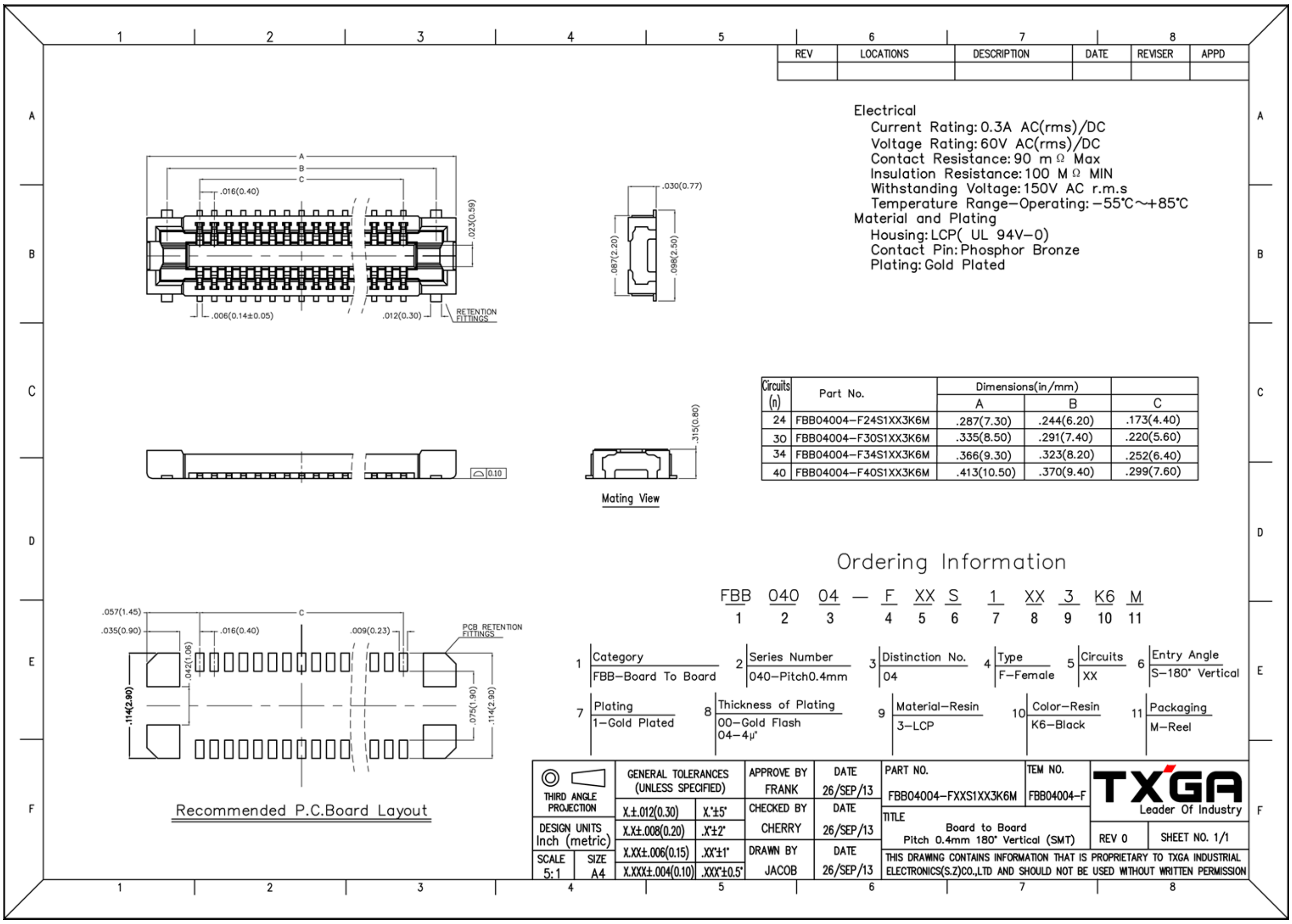

Figure 11: WisConnector PCB footprint and recommendations

Environmental Characteristics

The table below lists the operation and storage temperature requirements of RAK19010:

Parameter	Minimum	Typical	Maximum
Operational temperature range	-35 °C	+25 °C	+75 °C
Extended temperature range	-40 °C	+25 °C	+80 °C
Storage temperature range	-40 °C	+25 °C	+80 °C

Schematic Diagram

The component schematics diagram of the RAK19010 is shown in **Figure 12**.

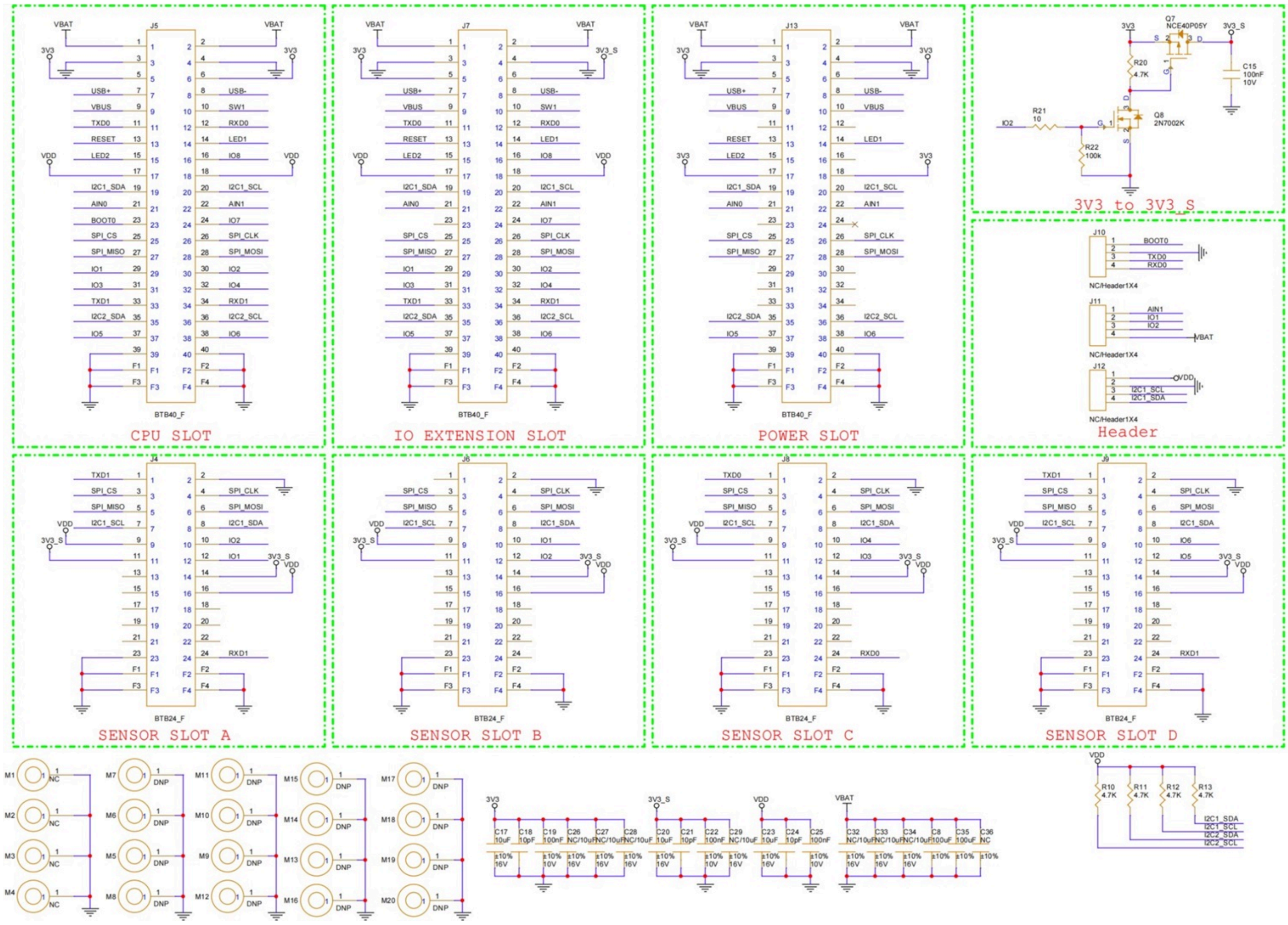

Figure 12: RAK19010 schematic diagram (Power)